

APPLICATION_FLOW.md

EduMentor AI

Application Flow Document

Version: 2.0

Status: MVP

Overview

This document describes the complete user journey and system workflows within EduMentor AI.

The application consists of:

- Authentication
- Onboarding
- Dashboard
- AI Tutor
- Adaptive Quizzes
- Progress Tracking
- Settings & Profile

All features are connected through a secure backend powered by Supabase and Groq.

High-Level User Flow

```
graph TD;
    A[Landing Page] --> B[Register / Login];
    B --> C[Onboarding];
    C --> D[Dashboard];
    D --> E[Learning Session];
    E --> F[AI Tutor];
```

↓
Adaptive Quiz
↓
Progress Update
↓
Continue Learning

Authentication Flow

User Registration

User
↓
Register Form
↓
Supabase Auth
↓
Account Created
↓
Profile Created
↓
Redirect to Onboarding

User Login

User
↓
Login Form
↓
Supabase Auth
↓
Session Created
↓
Dashboard

Protected Route Validation

Every protected page follows:

```
User Request
  ↓
Check Session
  ↓
Valid Session?
  ↓
  Yes → Continue
  ↓
  No  → Redirect Login
```

Protected Routes:

```
/dashboard
/dashboard/learn
/dashboard/quiz
/dashboard/progress
/profile
/settings
```

Onboarding Flow

Purpose:

Collect learning preferences.

User Journey

```
Onboarding Form
  ↓
Learning Topic
  ↓
Skill Level
  ↓
Learning Goal
  ↓
```

Daily Study Time



Save Preferences



Dashboard

Database Updates

Tables:

user_preferences

Data Stored:

- Learning Topic
- Skill Level
- Learning Goal
- Daily Study Time

Dashboard Flow

Purpose:

Provide centralized learning overview.

Dashboard Components

Continue Learning

Recent Sessions

Progress Overview

Weak Topics

Quick Actions

Dashboard Loading Flow

```
graph TD; A[User Opens Dashboard] --> B[Validate Session]; B --> C[Fetch Analytics]; C --> D[Fetch Recent Sessions]; D --> E[Render Dashboard];
```

Learning Session Flow

Purpose:

Provide persistent AI tutoring.

New Session Creation

```
graph TD; A[Create Session] --> B[Create Conversation Record]; B --> C[Open Chat Interface];
```

Resume Session

```
graph TD; A[Select Previous Session] --> B[Load Conversation]; B --> C[Load Messages]; C --> D[Continue Learning];
```

Rename Session

```
graph TD; A[Edit Session Title] --> B[Update Conversation];
```

Delete Session

```
graph TD; A[Delete Session] --> B[Delete Messages]; B --> C[Delete Conversation];
```

AI Tutor Flow

Purpose:

Provide personalized tutoring.

Message Flow

```
graph TD; A[User Sends Message] --> B[Store User Message]; B --> C[Build Context]; C --> D[Call Groq API]; D --> E[Receive Response]; E --> F[Store AI Response]; F --> G[Display Response];
```

Context Building Strategy

To avoid context overflow.

Only include:

Last 15 Messages

Current User Message

Future Expansion:

Top 5 Knowledge Chunks

Prompt Structure

System Prompt

+

Last 15 Messages

+

Current User Message

AI Architecture

Frontend

↓

POST /api/chat

↓

Server

↓

Groq

↓

Server

↓

Frontend

Chat Storage Flow

Every message is persisted.

```
graph TD;
    A[User Message] --> B[messages Table];
    C[AI Response] --> D[messages Table];
```

Full history remains available.

Rate Limiting Flow

Purpose:

Prevent abuse and API exhaustion.

Chat Request Validation

```
graph TD;
    A[Incoming Request] --> B[Check api_usage];
    B --> C[Requests Exceeded?];
    C -- Yes --> D[Block Request];
    C -- No --> E[Continue];
```

Limits

```
100 Chat Requests Per Day
```


per user.

Adaptive Quiz Flow

Purpose:

Generate personalized assessments.

Quiz Generation Workflow

```
graph TD; A[Generate Quiz] --> B[Retrieve Progress]; B --> C[Read Mastery Score]; C --> D[Determine Difficulty]; D --> E[Generate Quiz]; E --> F[Store Quiz]; F --> G[Display Quiz];
```

Difficulty Selection Rules

Mastery Score	Difficulty
0 - 40	Beginner
41 - 70	Intermediate
71 - 100	Advanced

Example

Mastery = 25
Difficulty = Beginner

Mastery = 85
Difficulty = Advanced

Quiz Submission Flow

Submit Quiz
↓
Evaluate Answers
↓
Calculate Score
↓
Store Results
↓
Update Progress
↓
Display Feedback

Quiz Storage Flow

Tables Used:

quizzes
quiz_results

Quiz Rate Limiting

Generate Quiz
↓

Check api_usage



Limit Exceeded?



Yes → Block

No → Continue

Limit:

50 Quiz Generations Per Day

Progress Tracking Flow

Purpose:

Measure learning growth.

Progress Update Trigger

Triggered after:

- Quiz Completion
 - Session Completion
-

Progress Workflow

Quiz Submitted



Calculate Performance



Update Mastery Score



Update Analytics



Refresh Dashboard

Mastery Calculation

Example Formula:

```
(  
Average Quiz Score × 0.7  
)  
+  
(  
Completion Consistency × 0.3  
)
```

Result:

0 - 100

Analytics Flow

Dashboard Analytics:

```
Total Sessions  
  
Total Messages  
  
Total Quizzes  
  
Average Score  
  
Learning Time
```

Analytics Update

```
User Activity  
  ↓  
Update learning_analytics  
  ↓  
Refresh Dashboard Widgets
```

Profile Flow

Route:

/profile

Features

Update Name

Update Avatar

View Email

Database

profiles

Settings Flow

Route:

/settings

Features

Theme Settings

Learning Preferences

Account Management

Security Flow

Purpose:

Protect user data.

Request Validation

Every protected request:

```
graph TD; Request --> ValidateSession[Validate Session]; ValidateSession --> VerifyUser[Verify User]; VerifyUser --> AuthorizeAccess[Authorize Access]; AuthorizeAccess --> ExecuteLogic[Execute Logic];
```

Row Level Security Flow

```
graph TD; UserRequest[User Request] --> RLSValidation[RLS Validation]; RLSValidation --> OwnData{Own Data?}; OwnData --> YesReturnData[Yes → Return Data]; OwnData --> NoDenyAccess[No → Deny Access];
```

Chat Privacy Flow

```
Fetch Messages
  ↓
Verify Conversation Owner
  ↓
Retrieve Messages
```

Users may never access another user's chat history.

API Key Security Flow

```
Browser
  ↓
API Route
  ↓
Groq
```

The browser never communicates directly with Groq.

Error Handling Flow

AI Failure

```
Groq Error
  ↓
Catch Error
  ↓
Return Friendly Message
```

Database Failure

```
Database Error
  ↓
Log Error
```

↓
Return Safe Response

Final System Flow

```
graph TD; Register --> Onboarding; Onboarding --> Dashboard; Dashboard --> CreateLearningSession[Create Learning Session]; CreateLearningSession --> ChatWithAITutor[Chat With AI Tutor]; ChatWithAITutor --> GenerateAdaptiveQuiz[Generate Adaptive Quiz]; GenerateAdaptiveQuiz --> SubmitQuiz[Submit Quiz]; SubmitQuiz --> UpdateProgress[Update Progress]; UpdateProgress --> ResumeLearningLater[Resume Learning Later];
```

MVP Success Flow

A new user should be able to:

1. Register an account.
2. Complete onboarding.
3. Start a learning session.
4. Chat with the AI tutor.
5. Generate an adaptive quiz.
6. Complete the quiz.
7. View progress analytics.
8. Resume learning at a later time.

If this workflow functions successfully, the MVP is considered complete.